Math Bootcamp

Day 3 — Linear algebra and real analysis

Math Camp 2026

How to use these slides

Main deck: definitions, recipes, examples, and practice.

Appendix and proofs (marked on relevant slides): **for interest only** — optional recap for those already comfortable with the math.

Not required for Math Camp. Focus on the main slides and exercises unless you want the extra detail.

3.1 Fixed points — equilibrium as a solution

Fixed point. A point x^* is a **fixed point** of a map T if

$$T(x^*) = x^*.$$

Economics reading. Many equilibria are fixed-point problems:

- **Market clearing:** price p^* such that excess demand $E(p^*) = 0$ (fixed point of an update rule or zero of a function).
- **Bellman equation:** value function V with $V = T(V)$ in dynamic programming.
- **Game theory:** Nash equilibrium as a fixed point of best-response maps.

Question. When does a fixed point **exist**? When is it **unique**?

3.1 Contraction Mapping Theorem

Setup. (X, d) a complete metric space; $T\colon X \to X$.

Contraction. T is a **contraction** if $\exists\, \alpha \in (0, 1)$ with

$$d\big(T(x), T(y)\big) \le \alpha\, d(x, y) \qquad \text{for all } x, y \in X.$$

Contraction Mapping Theorem (Banach). If T is a contraction on a complete metric space, then

- T has a **unique** fixed point x^*;
- iterating $x_{k+1} = T(x_k)$ converges to x^* from any start.

Intuition: T brings points closer together — at most one fixed point, and iteration finds it.

3.1 Brouwer Fixed Point Theorem

Brouwer (statement). If $K \subset \mathbb{R}^n$ is **nonempty, compact, and convex** and $T : K \to K$ is **continuous**, then T has at least one fixed point.

Low-dimensional intuition.

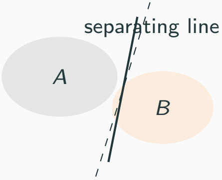
- $n = 1$: a continuous map from $[0, 1]$ to itself must cross the diagonal $y = x$.
- $n = 2$: stirring coffee in a cup — some molecule ends where it started.

Contrast with contraction. Brouwer gives **existence** (not uniqueness) under compactness, convexity, and continuity. Used when equilibrium maps are not contractions but stay inside a convex feasible set.

for interest only — proof uses advanced topology; we use the statement only.

3.2 Separating hyperplanes — picture

Separation (informal). If A and B are **disjoint convex** sets in $\mathbb{R}^n$, often we can find a **hyperplane** that puts A on one side and B on the other.



$$\exists\, \mathbf{p} \neq \mathbf{0},\ \beta \in \mathbb{R} : \quad \mathbf{p}\cdot a \leq \beta \leq \mathbf{p}\cdot b \quad \forall\, a \in A,\ b \in B.$$

Supporting hyperplane. At a boundary point of a convex set, a hyperplane can "touch" the set and leave the set on one side.

3.2 Why separation matters in economics

Welfare economics (high level).

- **Pareto efficiency:** under convexity, Pareto-improving reallocations are limited — efficient allocations often lie on boundaries where supporting prices separate feasible and preferred sets.
- **Prices as separators:** in competitive equilibrium, a price vector can support an allocation by separating consumers' upper contour sets from aggregate production possibilities.
- **Duality preview:** separating hyperplanes link **primal** problems (choices) to **dual** objects (shadow prices, multipliers) — more on Day 4–5.

Takeaway. Convexity + separation $\Rightarrow$ existence of linear prices that rationalize efficient allocations (statement-level; no proof required).